SiGMaR Installation Guide

Installation on macOS (.dmg), Windows (.exe) and Linux, with troubleshooting for each operating system.

macOS — .dmg	Windows — .exe	Linux — .deb

SiGMaR — Molecular Subtyping Dashboard
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1 Introduction and requirements

SiGMaR is a local dashboard for molecular subtyping analysis. It is not a cloud service: it runs entirely on the user's computer inside Docker containers, and is used from the browser at `http://localhost:8060`.

MANDATORY PREREQUISITE

Before installing SiGMaR, Docker must already be installed and running on the computer (Docker Desktop on macOS/Windows, Docker Engine on Linux). Without Docker running, the SiGMaR installer cannot load the application images and the program will not start. If Docker is not yet installed, the complete instructions for Windows, macOS, and Linux are in the Appendix at the end of this document.

Recommended minimum requirements

Minimum configuration

CPU	4 cores
RAM	8 GB
Free disk space	at least 30 GB
Suited for	testing and small datasets

Recommended configuration

CPU	6-8 cores
RAM	16 GB
Free disk space	50-70 GB
Suited for	full pipeline (DESeq2 + TMP)

WHY SO MUCH SPACE IS NEEDED

The Docker images included in the package take up several GB (the core app and the classification models bundle). During installation, Docker imports these images and saves their layers to disk: the actual space used can therefore exceed that of the installation folder alone.

2 Installing SiGMaR on macOS (.dmg)

BEFORE YOU BEGIN

Make sure Docker Desktop is installed and running (steady whale icon in the menu bar). If it is not yet installed, see the Appendix at the end of the document.

1. Download the provided `SiGMaR.dmg` file and double-click to open it. A window will open with the application installer.
2. Double-click `Install SiGMaR.app` to start the installation. If macOS shows a security warning ("app from an unidentified developer"), open System Settings → Privacy & Security and click "Open Anyway" next to the message about SiGMaR.
3. The installer copies the application files and loads the included Docker images (this can take several minutes on first run, as it involves several GB of data).
4. When finished, the installer creates the application icon on the Desktop: `~/Desktop/SiGMaR.app`
5. Double-click `SiGMaR.app` to launch the application. The launcher starts Docker (if not already running), starts the SiGMaR container, and automatically opens the browser at `http://localhost:8060`.

IF YOU RUN INTO PROBLEMS

If startup fails, the detailed log is located at:

```
~/Library/Logs/SiGMaR/launcher.log
```

Viewable from Terminal with:

```
tail -n 80 ~/Library/Logs/SiGMaR/launcher.log
```

For other specific symptoms, see chapter 6, "Common issues — macOS".

3 Installing SiGMaR on Windows (.exe)

BEFORE YOU BEGIN

Make sure Docker Desktop is installed and running (the icon in the taskbar shows “running”). If it is not yet installed, see the Appendix at the end of the document.

1. Download both files (`SiGMaR-Setup.exe` and `SiGMaR-Setup-1.bin`) into the same folder, for example the Downloads folder.
2. Double-click `SiGMaR-Setup.exe` to start the setup wizard. Installation takes place in the current user's profile and does not require administrator rights or a UAC confirmation dialog.
3. Follow the wizard (Next → Next → Install). By default, SiGMaR is installed in: `%LocalAppData%\Programs\SiGMaR`
4. During installation, the wizard checks that Docker is running and loads the included Docker images. This step is slow (it can take several minutes on first run, as it involves a few GB of images): wait for it to finish without closing the window.
5. When finished, the installer automatically creates a shortcut on the Desktop and closes with Finish.
6. Double-click the SiGMaR shortcut on the Desktop: the launcher starts the Docker container and opens the browser at `http://localhost:8060`.

WATCH OUT FOR THE FILES

The Windows package consists of two files that must stay in the same folder: `SiGMaR-Setup.exe` and `SiGMaR-Setup-1.bin`. The second contains part of the installation data (the Docker images exceed the limit of a single executable): if it is moved or deleted separately, the installation will fail.

IF DOCKER IS NOT READY

If an error related to Docker appears during installation or startup, manually open Docker Desktop from the Start menu, wait until it shows “running”, and retry the operation (relaunch the installer or the SiGMaR shortcut). For other specific symptoms, see chapter 7, “Common issues — Windows”.

4 Note for Linux (.deb)

On Linux (Debian/Ubuntu-based distributions) SiGMaR is distributed as the `sigmar_1.0_amd64.deb` package. After installing Docker Engine and Docker Compose (see Appendix):

```
sudo apt install ./sigmar_1.0_amd64.deb
```

The package installs the application entry in the system menu (`/usr/share/applications/`). For a quick icon on the Desktop:

```
cp /usr/share/applications/sigmar.desktop ~/Desktop/
chmod +x ~/Desktop/sigmar.desktop
```

On GNOME environments, after copying, you may need to right-click the file → Allow Launching, as unsigned launchers are blocked by default.

MORE DETAIL

Chapter 8, “Common issues — Linux”, collects the most frequent problems encountered during installation on Linux, with cause and solution for each.

5 First launch and shutdown

Regardless of operating system, once started, SiGMaR is used from the browser at the address:

```
http://localhost:8060
```

If the browser does not open automatically on first launch, manually open this address in Chrome, Safari, Firefox, or Edge.

To shut down the application, from Terminal/PowerShell in the SiGMaR installation folder:

```
docker compose down
```

Docker Desktop can remain running: SiGMaR will be restarted by double-clicking again the icon/shortcut created during installation.

6 Common issues — macOS

! “App from an unidentified developer” prevents opening

Cause: Gatekeeper blocks apps not signed/notarized by Apple.

Fix: System Settings → Privacy & Security → click “Open Anyway” next to the message about SiGMaR, then try opening Install SiGMaR.app again.

! The app does not start / the page does not load

Cause: Docker Desktop is not running, or the Docker engine has not finished starting yet.

Fix: open Docker from Applications and wait for the whale icon to become steady, then retry. Check the log:

```
tail -n 80 ~/Library/Logs/SiGMaR/launcher.log
```

! Docker Desktop is not installed where SiGMaR expects it

Cause: SiGMaR looks for Docker Desktop at the standard path `/Applications/Docker.app`. If it was installed elsewhere, the launcher cannot start it automatically.

Fix: reinstall Docker Desktop by dragging the app into Applications (the default path proposed by Docker's .dmg installer).

! The first launch is very slow

Cause: on the first launch Docker must finish loading the included images (several GB); subsequent launches are much faster.

Fix: wait for it to finish without closing the Terminal/launcher window.

7 Common issues — Windows

! Docker Desktop reports “WSL 2 installation is incomplete” or does not start

Cause: virtualization is disabled in the BIOS/UEFI, or the WSL2 components are not yet enabled in Windows.

Fix 1 — Check whether virtualization is enabled:

• Open Task Manager (Ctrl+Shift+Esc) → Performance tab → CPU: “Virtualization: Enabled” should appear in the bottom right.

• Alternatively, from PowerShell: `systeminfo` and look for the line “Virtualization Enabled In Firmware: Yes” in the Hyper-V requirements section.

Fix 2 — If it is disabled, enable it from the BIOS/UEFI:

• Restart the PC and enter the BIOS/UEFI (key varies by manufacturer: Del, F2, F10, or Esc during startup — check the boot screen or the PC manual).

• Look, usually under the Advanced or CPU Configuration section, for an entry called Intel VT-x / Intel Virtualization Technology (Intel CPU) or AMD-V / SVM Mode (AMD CPU).

• Set it to Enabled, save the changes, and exit (typically F10 → confirm).

Fix 3 — Enable the Windows features required for WSL2 (from PowerShell as Administrator):

```
wsl --install
```

If the command is not available, manually enable the two features and restart:

```
dism.exe /online /enable-feature /featurename:Microsoft-Windows-Subsystem-Linux /all /norestart
dism.exe /online /enable-feature /featurename:VirtualMachinePlatform /all /norestart
```

After the restart, set WSL2 as the default version:

```
wsl --set-default-version 2
```

Finally, reopen Docker Desktop and wait until it shows “running”.

! The installer does not start or freezes immediately

Cause: SiGMaR-Setup.exe and SiGMaR-Setup-1.bin are not in the same folder, or one of the two has been renamed.

Fix: re-download both files, making sure they stay together, without renaming them, in the same folder.

! Windows/antivirus blocks the downloaded file (“Windows protected your PC”)

Cause: SmartScreen or the antivirus flag an executable that is not widely recognized.

Fix: in the SmartScreen window click “More info” then “Run anyway”. If the antivirus quarantines SiGMaR-Setup-1.bin, add an exception for the download folder before reinstalling.

! Loading the Docker images is very slow or gets stuck

Cause: insufficient disk space, or resources (CPU/RAM) assigned to Docker Desktop too low.

Fix: make sure at least 30-50 GB are free; increase resources in Docker Desktop → Settings → Resources (recommended: 8-16 GB RAM, 4-8 cores); restart Docker Desktop and repeat the installation.

! Port 8060 is already in use

Cause: another program is already using port 8060.

Fix: check from PowerShell:

```
netstat -ano | findstr :8060
```

Close the process using the port (via the PID shown) or change the exposed port in the docker-compose.win.yaml file in the installation folder.

! Where to find the logs in case of error

The launcher log is located at:

```
%LOCALAPPDATA%\SiGMaR\launcher.log
```

Viewable from PowerShell with:

```
Get-Content "$env:LOCALAPPDATA\SiGMaR\launcher.log" -Tail 80
```

FIX AVAILABLE

The TMP subtyping method (based on several external Docker models) could fail to complete correctly on Windows due to a path issue between nested Docker containers. The problem has been identified and fixed, and the fix has been verified with a complete, successful run of all TMP models. The fix will be included starting with the next version of the Windows installer; if the app is already installed from a previous version, reinstall with the latest SiGMaR-Setup.exe once available to get it.

8 Common issues — Linux

! apt install shows a permission error during download/cache

Symptom: “Download is performed unsandboxed as root as file ... couldn't be accessed by user '_apt'”.

Cause: apt reads the local .deb as the system user _apt, which does not have permission to traverse folders with restricted permissions (e.g. ~/Desktop at 700). It is not blocking: the installation proceeds anyway.

Check:

```
dpkg -l | grep sigmar
```

If “ii sigmar 1.0 amd64” appears, the package is installed despite the warning.

Fix to avoid the warning:

```
cp ~/Desktop/sigmar_1.0_amd64.deb /tmp/
sudo apt install /tmp/sigmar_1.0_amd64.deb
```

! No icon created on the Desktop after installation

Cause: the .deb package installs the application entry in /usr/share/applications/ (system menu), not a direct shortcut on the Desktop.

```
cp /usr/share/applications/sigmar.desktop ~/Desktop/
chmod +x ~/Desktop/sigmar.desktop
```

On GNOME/Nautilus you may then need to right-click the file → Allow Launching, as unsigned launchers are blocked by default.

! Startup page does not open: “Access to the file was denied” on startup.html

Most common cause: browser installed as a snap (Firefox/Chromium via snap run in a confined environment that denies access to paths like ~/.local/state, even with correct permissions).

Check:

```
snap list 2>/dev/null | grep -iE "chromlfirefox"
```

Definitive fix: install a version of the browser not confined by snap, for example native Firefox:

```
sudo add-apt-repository ppa:mozillateam/ppa
sudo apt update
sudo apt install -t 'o=LP-PPA-mozillateam' firefox
```

or for Chromium:

```
sudo snap remove chromium
sudo apt install chromium-browser
```

! docker: permission denied on the Docker socket

Cause: the user is not part of the docker group.

```
sudo usermod -aG docker $USER
```

Then log out and back in (or restart) for the new group to take effect.

! Port 8060 is already in use

Symptom: the container starts but the browser does not show the dashboard, or docker compose up reports a port conflict.

Check:

```
sudo ss -ltnp | grep 8060
```

Fix: close the process using the port, or change the exposed port in the project's docker-compose.yaml.

QUICK DIAGNOSIS SUMMARY

```
dpkg -l | grep sigmar                    # package installed?
docker info                              # Docker daemon running?
docker ps                                # SiGMaR container running?
snap list | grep -iE "chrom|firefox"     # browser sandboxed?
namei -l ~/.local/state/sigmar/startup.html  # permission chain ok?
```

A Appendix — Installing Docker

Docker is the engine that runs SiGMaR's containers. It must be installed before SiGMaR and must be running (the whale icon “running”) every time you want to use the app. Below are the instructions for the three operating systems.

REFERENCE SUPPLEMENT

Only needed if Docker is not yet installed on the computer: SiGMaR requires Docker to already be present and running before proceeding with chapters 2-4.

Windows

1. System requirement: 64-bit Windows 10/11 with virtualization enabled in the BIOS/UEFI. Docker Desktop requires the WSL2 subsystem.

   To check whether virtualization is already enabled: Task Manager (Ctrl+Shift+Esc) → Performance tab → CPU → check the “Virtualization” entry in the bottom right.

   If it is disabled: restart the PC, enter the BIOS/UEFI (Del, F2, F10, or Esc key depending on the manufacturer), find the Intel VT-x / Virtualization Technology (Intel) or AMD-V / SVM Mode (AMD) entry — usually under “Advanced” or “CPU Configuration” — set it to Enabled, then save and exit.

   Finally, from PowerShell as Administrator, enable WSL2:

   ```
   wsl --install
   ```

   (requires a PC restart when finished). Full details, including the manual fix via dism.exe, are in chapter 7 “Common issues — Windows”.

2. Download Docker Desktop for Windows from the official website: `https://www.docker.com/products/docker-desktop/`
3. Run Docker Desktop Installer.exe and follow the wizard (recommended WSL2 option, left selected by default). Restart the computer when finished if prompted.
4. Start Docker Desktop from the Start menu and wait until the icon in the taskbar shows “Docker Desktop is running”.
5. Verify the installation by opening PowerShell:

   ```
   docker --version
   docker info
   ```

If both commands respond without errors, Docker is ready.

macOS

1. Download Docker Desktop for Mac from the official website, choosing the correct version for your processor:

   `https://www.docker.com/products/docker-desktop/`

   — Apple Silicon (M1/M2/M3/M4) or Intel chip, depending on the Mac in use (can be checked via Apple Menu → About This Mac).
2. Open the downloaded .dmg file and drag the Docker icon into the Applications folder.
3. Launch Docker from Applications: on first launch macOS will ask for confirmation and may require the user password to complete the installation of network/virtualization components.
4. Wait for the whale icon in the menu bar to become steady (not animated): this means the Docker engine is running.
5. Verify from Terminal:

   ```
   docker --version
   docker info
   ```

IMPORTANT PATH FOR SIGMAR

On macOS, SiGMaR expects Docker Desktop to be installed at the standard path `/Applications/Docker.app`: this is the path created automatically by dragging the app into Applications in step 2. If Docker is installed elsewhere, SiGMaR's launcher may not be able to start it automatically.

Linux

1. On Debian/Ubuntu distributions, install Docker Engine and the Docker Compose plugin using the official convenience script:

   ```
   curl -fsSL https://get.docker.com -o get-docker.sh
   sudo sh get-docker.sh
   ```

   Alternatively, follow the apt-based procedure described at docs.docker.com/engine/install/ for your distribution.
2. Add your user to the docker group, to be able to run commands without sudo:

```
sudo usermod -aG docker $USER
newgrp docker
```

After this command you may need to log out and log back in (or restart) for the new group to take effect.

3. Enable and start the Docker service:

```
sudo systemctl enable --now docker
```

4. Verify the installation:

```
docker info
docker compose version
```

Both commands must respond without errors before installing SiGMaR.